A NOTE ON THE DE BRUIJN-SPRINGER INEQUALITY BETWEEN THE ZEROS AND CRITICAL POINTS OF POLYNOMIALS

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Abstract. The De Bruijn-Springer inequality was a long-standing conjecture, formulated in 1947, concerning the zeros and critical points of a complex polynomial. It was proved independently by Malamud and Pereira in 2003. Using their main result, we establish an aggregate weak majorization relationship between the zeros and critical points. This allows us to show that De Bruijn-Springer inequality is the last one in a sequence of inequalities, placing it into a broader geometric context. As a corollary, we derive a result that is complementary to a majorization relationship discovered by Schmeisser in 2004. Even though our corollary is stronger than Schmeisser's result, it applies to a narrower class of functions.

1. Introduction

Let

$$f(z) = a\prod_{i=1}^{n}(z - z_i), \quad a \neq 0,$$

be a complex polynomial of degree $n \geq 2$, with derivative

$$f'(z) = an\prod_{j=1}^{n-1}(z - w_j).$$

In [3], De Bruijn and Springer asked whether the inequality

$$\frac{1}{n-1}\sum_{i=1}^{n-1}\phi(w_i) \leq \frac{1}{n}\sum_{i=1}^{n}\phi(z_i)$$

holds for every convex function $\phi : \mathbb{C} \to \mathbb{R}$. This conjecture was proved independently by Malamud [6] and Pereira [8] (see also [5] for a survey on the applications of majorization theory to the geometric theory of polynomials).

We say that, for two vectors $x, y \in \mathbb{R}^n$, y *weakly majorizes* x, denoted $x \prec_w y$, if

$$\sum_{i=1}^{k} x_i^{\downarrow} \leq \sum_{i=1}^{k} y_i^{\downarrow}, \qquad k = 1, \ldots, n,$$

where $x^{\downarrow}$ denotes the non-increasing rearrangement of x.

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The purpose of this note is to point out the following extension of the De Bruijn–Springer inequality. For any convex function $\phi : \mathbb{C} \to \mathbb{R}$, let

$$\mathcal{Z}_\phi^{[n-1]} := (\underbrace{\phi(z_1), \dots, \phi(z_1)}_{n-1}, \dots, \underbrace{\phi(z_n), \dots, \phi(z_n)}_{n-1})^T \in \mathbb{R}^{n(n-1)} \tag{1}$$

and

$$\mathcal{W}_\phi^{[n]} := (\underbrace{\phi(w_1), \dots, \phi(w_1)}_{n}, \dots, \underbrace{\phi(w_{n-1}), \dots, \phi(w_{n-1})}_{n})^T \in \mathbb{R}^{n(n-1)}. \tag{2}$$

We show that

$$\mathcal{W}_\phi^{[n]} \prec_w \mathcal{Z}_\phi^{[n-1]}. \tag{3}$$

The weak majorization (3) yields an entire family of inequalities between the partial sums, with the last one being the De Bruijn–Springer inequality. This places the latter result into a broader geometric context: it is the endpoint of a sequence of inequalities between the zeros and the critical points. The proof is based on a majorization relationship between the zeros and the critical points discovered independently by Malamud and Pereira.

Relationship (3) also allows us to derive a counterpart to the following result established by Schmeisser [9, Corollary 3].

Theorem 1.1 (Schmeisser)**.** *Let f be a complex polynomial of degree n, with zeros $z_1, \dots, z_n$ and critical points $w_1, \dots, w_{n-1}$. Let $\psi : [0, \infty) \to \mathbb{R} \cup \{-\infty\}$ be such that $\psi \circ \exp$ is convex on $\mathbb{R} \cup \{-\infty\}$. Then*

$$(\psi(|w_1|), \dots, \psi(|w_{n-1}|), \psi(0)) \prec_w (\psi(|z_1|), \dots, \psi(|z_n|)). \tag{4}$$

This theorem refines the Gauss–Lucas theorem, since the latter is equivalent to the first majorization inequality in (4) with $\psi(x) = x$.

While Schmeisser's result applies to a broader class of functions, our framework yields a tighter majorization bound, see Corollary 3.3.

2. Background results

We begin with an elementary fact that will be used in the proof of the main theorem. It is a standard result that can be found in [1, Theorem II.1.3], among other places.

Lemma 2.1. *Let $x, y \in \mathbb{R}^n$. Then $x \prec_w y$ if and only if*

$$\sum_{i=1}^n (x_i - t)_+ \le \sum_{i=1}^n (y_i - t)_+, \ \text{for all } t \in \mathbb{R},$$

where $(s)_+ := \max\{s, 0\}$.

Proposition 2.2. *Let $z_1, \dots, z_n$ and $w_1, \dots, w_m$ be points in a real vector space, where $1 \le m \le n$. Assume that there exists an $m \times n$ matrix $A = (a_{ij})$ such that*

$$a_{ij} \ge 0, \ \sum_{j=1}^n a_{ij} = 1, \ \sum_{i=1}^m a_{ij} = \frac{m}{n},$$

and

$$w_i = \sum_{j=1}^n a_{ij} z_j, \ \text{for } i = 1, \dots, m.$$

Define the repeated vectors

$$z^{[m]} := (\underbrace{z_1, \ldots, z_1}_{m}, \ldots, \underbrace{z_n, \ldots, z_n}_{m})$$

and

$$w^{[n]} := (\underbrace{w_1, \ldots, w_1}_{n}, \ldots, \underbrace{w_m, \ldots, w_m}_{n}).$$

Then there exists a doubly stochastic matrix $D \in \mathbb{R}^{mn \times mn}$*, such that*

$$w^{[n]} = Dz^{[m]}. \tag{5}$$

Proof. Index the rows of D by the pairs (i,s), where $i = 1, \ldots, m$ and $s = 1, \ldots, n$. Index the columns by the pairs (j,t), where $j = 1, \ldots, n$ and $t = 1, \ldots, m$. Define

$$D_{(i,s),(j,t)} := \frac{a_{ij}}{m}.$$

For each row (i,s), we have

$$\sum_{j=1}^{n} \sum_{t=1}^{m} D_{(i,s),(j,t)} = \sum_{j=1}^{n} \sum_{t=1}^{m} \frac{a_{ij}}{m} = \sum_{j=1}^{n} a_{ij} = 1.$$

For each column (j,t), we have

$$\sum_{i=1}^{m} \sum_{s=1}^{n} D_{(i,s),(j,t)} = \sum_{i=1}^{m} \sum_{s=1}^{n} \frac{a_{ij}}{m} = \frac{n}{m} \sum_{i=1}^{m} a_{ij} = 1.$$

Thus D is doubly stochastic. Assuming the indexing $z^{[m]}_{(j,t)} = z_j$, the (i,s) component of $Dz^{[m]}$ is

$$\sum_{(j,t)} D_{(i,s),(j,t)} z^{[m]}_{(j,t)} = \sum_{j=1}^{n} \sum_{t=1}^{m} \frac{a_{ij}}{m} z_j = \sum_{j=1}^{n} a_{ij} z_j = w_i,$$

which proves (5). □

The following corollary is standard in multivariate majorization, see [7, Proposition 15.A.4] or the original work by Karlin and Rinott [4, Theorem 2].

Corollary 2.3. *Under the conditions in Proposition 2.2, for every convex function* $\phi : \mathbb{C} \to \mathbb{R}$*, we have*

$$n \sum_{i=1}^{m} \phi(w_i) \le m \sum_{i=1}^{n} \phi(z_i).$$

We also need the following majorization result discovered independently by Malamud [6, Theorem 4.7] and Pereira [8, Theorem 5.4].

Theorem 2.4. *Let* $z_1, \ldots, z_n$ *and* $w_1, \ldots, w_{n-1}$ *be the zeros and critical points of the polynomial* $f(z)$ *of degree* $n > 1$*. Then there exists an* $(n-1) \times n$ *matrix* $A = (a_{ij})$*, such that*

$$a_{ij} \ge 0, \quad \sum_{j=1}^{n} a_{ij} = 1, \quad \sum_{i=1}^{n-1} a_{ij} = \frac{n-1}{n},$$

and

$$w_i = \sum_{j=1}^{n} a_{ij} z_j, \quad \text{for } i = 1, \ldots, n-1.$$

3. The main result: aggregate majorization

Below is the main result of this note.

Theorem 3.1. *Let $z_1, \dots, z_n$ and $w_1, \dots, w_{n-1}$ be the zeros and critical points of the polynomial $f(z)$ of degree n. For any convex function $\phi : \mathbb{C} \to \mathbb{R}$, we have*

$$\mathcal{W}_\phi^{[n]} \prec_w \mathcal{Z}_\phi^{[n-1]},$$

where vectors $\mathcal{Z}_\phi^{[n-1]}$ and $\mathcal{W}_\phi^{[n]}$ are defined in (1) and (2).

Proof. Let $A = (a_{ij})$ be the $(n-1) \times n$ matrix, whose existence is guaranteed by Theorem 2.4. By Proposition 2.2, there is a doubly stochastic matrix $D \in \mathbb{R}^{(n-1)n \times (n-1)n}$, such that

$$w^{[n]} = Dz^{[n-1]}.$$

For any fixed $t \in \mathbb{R}$, consider the function

$$\Phi_t(\xi) := (\phi(\xi) - t)_+.$$

Since $s \mapsto (s-t)_+$ is a convex non-decreasing function on $\mathbb{R}$ and ϕ is convex on $\mathbb{C}$, their composition is a convex function on $\mathbb{C}$, see [2, Section 3.2.4]. Applying Corollary 2.3 with the convex function Φ_t, we obtain

$$n \sum_{j=1}^{n-1} (\phi(w_j) - t)_+ \leq (n-1) \sum_{i=1}^{n} (\phi(z_i) - t)_+, \qquad \text{for all } t \in \mathbb{R}.$$

The conclusion follows from Lemma 2.1. $\square$

Theorem 3.1 yields a few corollaries. Let $\phi(z) := (\phi(z_1), \dots, \phi(z_n))$ and $\phi(w) := (\phi(w_1), \dots, \phi(w_{n-1}))$, and let $\phi(z)^\downarrow$ and $\phi(w)^\downarrow$ denote the non-increasing rearrangements of these vectors.

Corollary 3.2. *Let f be a complex polynomial of degree n, with zeros $z_1, \dots, z_n$ and critical points $w_1, \dots, w_{n-1}$. Let $\phi : \mathbb{C} \to \mathbb{R}$ be a convex function. Then*

$$\sum_{i=1}^{k} \phi(w)_i^\downarrow \leq \frac{n-1}{n} \sum_{i=1}^{k} \phi(z)_i^\downarrow + \frac{k}{n} \phi(z)_{k+1}^\downarrow, \qquad k = 1, \dots, n-1. \tag{6}$$

Proof. Theorem 3.1 implies that the sum of the largest nk entries of $\mathcal{W}_\phi^{[n]}$ is bounded above by the sum of the largest nk entries of $\mathcal{Z}_\phi^{[n-1]}$. The largest nk entries of $\mathcal{W}_\phi^{[n]}$ consist of n copies of each of the largest k values $\phi(w)_1^\downarrow, \dots, \phi(w)_k^\downarrow$, that is

$$\sum_{\alpha=1}^{nk} (\mathcal{W}_\phi^{[n]})_\alpha^\downarrow = n \sum_{i=1}^{k} \phi(w)_i^\downarrow.$$

On the other hand, the largest nk entries of $\mathcal{Z}_\phi^{[n-1]}$ consist of $\phi(z)_1^\downarrow, \dots, \phi(z)_k^\downarrow$, each repeated $n-1$ times, followed by exactly k copies of $\phi(z)_{k+1}^\downarrow$. Hence, their sum is

$$\sum_{\alpha=1}^{nk} (\mathcal{Z}_\phi^{[n-1]})_\alpha^\downarrow = (n-1) \sum_{i=1}^{k} \phi(z)_i^\downarrow + k\phi(z)_{k+1}^\downarrow.$$

The inequality follows. $\square$

In the case when $f(x) = \prod_{i=1}^{n}(x - \lambda_i)$ is a polynomial with real roots $\lambda_1 \geq \cdots \geq \lambda_n$, its critical points $\mu_1 \geq \cdots \geq \mu_{n-1}$ interlace with the roots. Then Theorem 3.1 and Corollary 3.2, applied to the convex functions $\phi(z) = \pm\mathrm{Re}(z)$, give the bounds

$$\sum_{j=1}^{k} \mu_j \leq \frac{n-1}{n} \sum_{i=1}^{k} \lambda_i + \frac{k}{n}\lambda_{k+1} \text{ and } \sum_{j=n-k}^{n-1} \mu_j \geq \frac{n-1}{n} \sum_{i=n-k+1}^{n} \lambda_i + \frac{k}{n}\lambda_{n-k},$$

for any $k = 1, \ldots, n-1$.

For analogous aggregate bounds concerning the interlacing eigenvalues of Hermitian matrices and their principal submatrices, refer to [10].

Corollary 3.3. *Let f be a complex polynomial of degree n, with zeros $z_1, \ldots, z_n$ and critical points $w_1, \ldots, w_{n-1}$. Let $\phi : \mathbb{C} \to \mathbb{R}$ be a convex function. Then*

$$(\phi(w_1), \ldots, \phi(w_{n-1}), c) \prec_w (\phi(z_1), \ldots, \phi(z_n)),$$

for any constant $c \leq \frac{1}{n}\sum_{i=1}^{n} \phi(z_i)$. In particular, this holds for $c = \min_{1 \leq i \leq n} \phi(z_i)$.

Proof. The k largest elements of the augmented vector $(\phi(w_1), \ldots, \phi(w_{n-1}), c)$ take one of two forms: they are either the k largest elements of $(\phi(w_1),\ldots,\phi(w_{n-1}))$, or they consist of c along with the $k-1$ largest elements of $(\phi(w_1), \ldots, \phi(w_{n-1}))$. Thus, we consider two cases.

Case 1: The sum of the k largest elements is $\sum_{j=1}^{k} \phi(w)^{\downarrow}_j$, which implies $k \leq n-1$. Since $\phi(z)^{\downarrow}$ is sorted in non-increasing order, we have $k\phi(z)^{\downarrow}_{k+1} \leq \sum_{i=1}^{k} \phi(z)^{\downarrow}_i$. Substituting this into (6) yields

$$\sum_{i=1}^{k} \phi(w)^{\downarrow}_i \leq \frac{n-1}{n} \sum_{i=1}^{k} \phi(z)^{\downarrow}_i + \frac{k}{n}\phi(z)^{\downarrow}_{k+1} \leq \sum_{i=1}^{k} \phi(z)^{\downarrow}_i.$$

Case 2: The sum of the k largest elements includes c, meaning it is of the form $c + \sum_{j=1}^{k-1} \phi(w)^{\downarrow}_j$. Using the hypothesis

$$c \leq \frac{1}{n} \sum_{i=1}^{n} \phi(z_i) = \frac{1}{n} \sum_{i=1}^{n} \phi(z)^{\downarrow}_i,$$

together with inequality (6), with $k-1$, we obtain

$$\begin{aligned} c + \sum_{i=1}^{k-1} \phi(w)^{\downarrow}_i &\leq \frac{1}{n} \sum_{i=1}^{n} \phi(z)^{\downarrow}_i + \frac{n-1}{n} \sum_{i=1}^{k-1} \phi(z)^{\downarrow}_i + \frac{k-1}{n}\phi(z)^{\downarrow}_k \\ &= \frac{1}{n}\Big(\sum_{i=1}^{k-1} \phi(z)^{\downarrow}_i + \phi(z)^{\downarrow}_k + \sum_{i=k+1}^{n} \phi(z)^{\downarrow}_i \Big) \\ &\quad + \frac{n-1}{n} \sum_{i=1}^{k-1} \phi(z)^{\downarrow}_i + \frac{k-1}{n}\phi(z)^{\downarrow}_k \\ &= \sum_{i=1}^{k-1} \phi(z)^{\downarrow}_i + \frac{k}{n}\phi(z)^{\downarrow}_k + \frac{1}{n} \sum_{i=k+1}^{n} \phi(z)^{\downarrow}_i. \end{aligned} \tag{7}$$

Consider the trivial identity

$$\phi(z)^{\downarrow}_k - \frac{k}{n}\phi(z)^{\downarrow}_k - \frac{1}{n} \sum_{i=k+1}^{n} \phi(z)^{\downarrow}_i = \frac{1}{n} \sum_{i=k+1}^{n} \Big(\phi(z)^{\downarrow}_k - \phi(z)^{\downarrow}_i \Big) \geq 0,$$

where the inequality holds because $\phi(z)^{\downarrow}_k \geq \phi(z)^{\downarrow}_i$ for all $i \geq k+1$. Adding the left-hand side of the last displayed identity to (7) shows that

$$c + \sum_{i=1}^{k-1} \phi(w)^{\downarrow}_i \leq \sum_{i=1}^{k} \phi(z)^{\downarrow}_i,$$

concluding the proof. □

Corollary 3.4. *Let f be a complex polynomial of degree n, with zeros $z_1, \dots, z_n$ and critical points $w_1, \dots, w_{n-1}$. Let $\psi : [0, \infty) \to \mathbb{R}$ be a non-decreasing convex function. Then*

$$(\psi(|w_1|), \dots, \psi(|w_{n-1}|), c) \prec_w (\psi(|z_1|), \dots, \psi(|z_n|)), \tag{8}$$

for any constant $c \leq \frac{1}{n}\sum_{i=1}^{n} \psi(|z_i|)$. In particular, this holds for $c = \psi(0)$.

Proof. Consider the function $\phi : \mathbb{C} \to \mathbb{R}$ defined by $\phi(z) = \psi(|z|)$. Since the modulus function $z \mapsto |z|$ is convex on $\mathbb{C}$ and ψ is a non-decreasing convex function on $[0, \infty)$, their composition ϕ is convex on $\mathbb{C}$ (see [2, Section 3.2.4]). Finally, note that $\psi(0) \leq \min_{1 \leq i \leq n} \psi(|z_i|)$.

□

Remark 3.5. Corollary 3.4 provides a stronger version of Schmeisser's result [9, Corollary 3], albeit for a slightly narrower class of functions. Schmeisser proved the weak majorization

$$(\psi(|w_1|), \dots, \psi(|w_{n-1}|), \psi(0)) \prec_w (\psi(|z_1|), \dots, \psi(|z_n|))$$

under the assumption that $\psi \circ \exp$ is a convex function on $\mathbb{R} \cup \{-\infty\}$.

Our assumption in Corollary 3.4 restricts ψ to the class of non-decreasing convex functions on $[0, \infty)$. This is a proper subclass of the functions considered by Schmeisser. To see this, note that the composition of a non-decreasing convex function ψ with the convex function exp is always convex. The inclusion is strict: for instance, $\psi(x) = \sqrt{x}$ satisfies Schmeisser's condition but is concave, hence excluded from our class.

Despite this slightly narrower scope, our conclusion is stronger. While Schmeisser appends the fixed term $\psi(0)$ to the critical points, we can append any constant $c \leq \frac{1}{n}\sum_{i=1}^{n} \psi(|z_i|)$. Because ψ is non-decreasing (a property that is implied by Schmeisser's assumption), we have $\psi(0) \leq \frac{1}{n}\sum_{i=1}^{n} \psi(|z_i|)$. Thus, selecting $c = \psi(0)$ recovers Schmeisser's exact bound, whereas choosing a larger c yields a tighter majorization relationship.

The next example shows that one cannot replace $\psi(0)$ with $\frac{1}{n}\sum_{i=1}^{n} \psi(|z_i|)$ in Schmeisser's result.

Example 3.6. Consider the polynomial $f(z) = z^2 - 3z + 2$ with zeros $z_1 = 2, z_2 = 1$ and critical point $w_1 = 1.5$. The function $\psi(x) = \log x$ satisfies Schmeisser's condition, but it is not convex on $[0, \infty)$. If our strengthened majorization (8) were to hold for $\psi(x) = \log x$ with the average $c = \frac{1}{2}(\log 2 + \log 1) = \frac{1}{2}\log 2$, the sum of all elements in the weakly majorized vector would be $\log 1.5 + \frac{1}{2}\log 2 \approx 0.752$. This strictly exceeds the sum of all elements in the majorizing vector $\log 2 + \log 1 \approx 0.693$, violating the weak majorization.

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